Available online at www.sciencedirect.com

ScienceDirect

IFAC PapersOnLine 58-8 (2024) 306–311

Circular product design for allowing re-using and re-purposing of products and components: a conceptual framework for circularity

Foivos Psarommatis*, **, Victor Azamfierei*, John David Lindström ****

* *University of Oslo, SIRIUS labs, Gaustadalleen 23B, 0373, Oslo, Norway (e-mail: foivosp@ifi.uio.no)*

** *Zerofect, Zug, Switzerland (fp@zerofect.com)*

*** *Politecnico di Milano, Italy (e-mail:victor.azamfirei@mdu.se)*

****Research Institutes of Sweden (RISE), Aurorum 2, 977 75 Luleå, Sweeden, (e-mail: john.david.lindstrom@ri.se)*

Abstract: We live in a fast-consuming era, where manufacturers producing products faster than ever and with only one lifecyle in mind. This phenomena has resulted in products being designed and manufactured with attention to the company profit and insufficient attention to the implementation of circular approached and by extent sustainability. Currently, the implementation of circularity is only considered at the end of the life of a product and not from the beginning during the design of the product. To achieve circularity there is the need to add more requirements to the product design that are coming from the concept of circular economy. Therefore, in the current paper we have defined additional design steps that are adding more requirement. Those steps include the incorporation of the product re-use, repurpose, re-manufacture and re-cycling. We provide a structural approach to quantifying circularity in design in order to close the gap between conventional design methods and sustainability.

Copyright © 2024 The Authors. This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0/)

Keywords: Circularity, Circular Design, Sustainability, Re-use, Re-purpose, Re-manufacture, Product Desing.

1. INTRODUCTION

Today manufacturing industry is facing unprecedented high environmental as well as customer requirements. One example is the goal set by the European Commission (EC) to reach carbon neutrality by 2050 through the sustainable and digital transformation (Industry 5.0 | European Commission 2021). The highly anticipated Circular Economy (CE) business model allows for achieving economic growth without depletion of resources and environmental degradation (Cong *et al.* 2019). CE is an economic system based on the recycling and of materials or products, particularly to maintain manufacturing in a sustainable way (Psarommatis and May 2022). Specific and well-established contributions in the CE domain derive from design, material science, and additive manufacturing (Sassanelli *et al.* 2020, Romani *et al.* 2021, Urbinati *et al.* 2023). Product Design underpins all three circular economy principles (i) waste elimination, (ii) products and materials circulation, and (iii) being regenerative by nature. Design practices/disciplines such as “Product Design”, “Engineering Design”, or “Industrial Design” to name a few are believed to impact between 70 to 80% of downstream activities such as manufacture and assembly (Cong *et al.* 2019). Eco-Design and Design for X (DfX) approaches are fundamental for the reduction of the negative impact. While one focuses on the material selection the other approach focuses on the manufacturing process.

The life cycle of a product is characterised by four major phases (i) procurement of ‘resources’, (ii) ‘manufacturing’ of the product according to the developed product designs, (iii) ‘usage’, and (iv) disposal at the ‘end-of-life’ (Ulrich and Eppinger 2016, K.e.k *et al.* 2021, Psarommatis and May 2022). Product Lifecycle Management (PLM) is the process of managing the entire lifecycle of a product from its engineering, design and manufacture to its service and disposal. PLM integrates people, data, processes, and business systems and provides a product information foundation for companies and their extended enterprises (Terzi *et al.* 2010, McKendry *et al.* 2022). Fuzzy design requirements introduced with CE increase the uncertainty of the design results (Guo *et al.* 2021). Digital Product Passport (DPP) can serve as an enabling technology to sustainability and circularity (European Commission, 2020a). DPP can store data from the entire lifecycle of the product, from the manufacturing, usage and maintenance, according to (Psarommatis and May 2024). All those data can be used for the efficient management of the used products and enabling circular approaches such us product reuse, repurpose and recycling. Additionally the DPP can assist on the information traceability and transparency (Acerbi *et al.* 2022, King *et al.* 2023, Psarommatis and May 2024).

The contemporary problem that many companies face when transitioning towards CE is characterised as multi-dimensional since it involves not only one company’s people, processes, and technologies used but of all the companies in one circular loop (Bocken *et al.* 2014). Design is the cradle of such chain and to secure a cradle to cradle (re-manufacturing, re-use, or re-purpose), the concept of Circular Product Design needs to be properly formulated. The aim of this paper is to consider

2405-8963 Copyright © 2024 The Authors. This is an open access article under the CC BY-NC-ND license.
Peer review under responsibility of International Federation of Automatic Control.
10.1016/j.ifacol.2024.08.138

such limitations and propose a Circular Product Design framework for it align to CE and allow companies to perform more efficiently and in sustainable manner asset lifecycle management (Jabbour *et al.* 2023).

The paper is organised as follows. Section 2 presents the state of the art of sustainable product design for CE. Section 3 analyses the proposed design framework for circularity. Section 4 critically discusses the proposed framework. Finally, Section 5 concludes the major contribution and describes promising future research topics in this area.

2. STATE OF THE ART

Product Design is vital for the implementation of efficient CE and is the first step in PLM. Contributions are organised in three categories (i) Circular Product Design Principles, (ii) Human centricity in Circular Product Design as true sustainability cannot be reached without the human aspect, and (iii) key design tools for achieving Circular Product Design, where DPP for PLM is deeply discussed. The selection of those three categories is based on authors' perspective and knowledge. To achieve CE it is imperative to have circular design and the tools to achieve it, but also the human aspect plays a crucial role to this scope.

Concerning a wanted longer usage phase of products or further advanced offerings, there need to be a design enabling ability to maintain/update/upgrade the constituents, such as hardware, software, services/support, or management of operations, etc., through-out the lifecycle to continue to deliver the necessary value to the customer (or the following customer if down-cycling, re-use, or re-purposing, in another context are applicable) and provider (Psarommatis and May 2022). All these constituents may have their own sub-lifecycle, which needs to be managed according to the business model and contract type used through the primary, secondary or further lifecycles following (Rentizelas and Trivyza 2022). Thus, there are a number of eco-innovations and eco-efficiency practices which can be selected and applied depending on context and product: re-cyclability, maintainability, upgradability, changeability, durability, cybersecurity, or reliability/availability, etc. In addition, there are several applicable design patterns and best practices that can be used as well – and just having the adequate knowledge and competencies can save a lot of time/effort instead of re-inventing what already exists. Recycling and waste generation by the proses of recycling, why we need more circular methods as well as design to enable long and extendable use-phases of the lifecycles.

2.1 Circular Product Design Principles

There is a lack of understanding on the type and amount of data that should be included within a DPP for a given application (King *et al.* 2023). According to Jansen et al. (2023) the main goals of DPP are to improve the circularity of products in terms of the R-Strategies and to stimulate transparency and traceability of products, materials, and components (Jansen *et al.* 2023). The R-Strategies of product sustainability management can be included in the CE framework to assess the product viability (K.e.k *et al.* 2021), a)Re-use used component(s) in the process, b) Reduce the procedures of overall materials/time consumption rates, c) Re-cycle materials/components, d)Recondition or Repurpose materials/components to suit a particular purpose, e) Recover any by-product/potentially usable material for procurement/treatment and involvement in further product development processes, f) Repair machine tools/tools of work, g)Remanufacture or remaking of some tools which have a short lifespan, h)Responsibility or Risk that is put into manufacturing/procuring materials for that product.

2.2 Human centricity in Circular Product Design

Product Design affects and is affected by the company's knowledge, processes, and available manufacturing systems as well as the global market (Tolio *et al.* 2017). Product Data such as composition, material employed, and required manufacturing process are vital for the assessment of hazardousness level to human health or to the environment (Jansen *et al.* 2023). Further, King et al. (2023) provided perspectives on a system for product impact assessment using DPP; in which the social impact of Product Designs consisted of (i) minimum social and labour condition, (ii) consumer rights, (iii) rights of indigenous people, (iv) keeping condition of animals (King *et al.* 2023).

2.3 Key design tools for Circular Product Design

The engineering design process is characterised the following iterative and fundamental steps i.e., (i) defining the problem, (ii) imagining or conceptualizing potential solutions, (iii) prototyping and testing, and (iv) improving the design. In the following subsection, the shortcomings of each of the design steps for CE are listed. **Defining the problem:** Specific and well-established contributions in the CE domain derive from design, material science, and additive manufacturing; however, their interdisciplinary relationship in CE contexts is relatively unexplored (Romani *et al.* 2021). **Conceptualisation of potential solutions:** Concept design is a critical step in creating innovative, complex and human-centric products. Cong et al. (2019) provided a design method to improve product recyclability by determining by optimising product end-of-use component flows (Cong *et al.* 2019). Guo et al. (2021) provided a smart knowledge deployment method is proposed for reasoning, configuring, and optimizing the conceptual design based on carbon emission evaluation and interaction (Guo *et al.* 2021). **Prototype and testing:** Shevchenko & Cluzel (2023) provided a CPD Toolkit for (i) diagnosability of the R-Strategies and (ii) appropriate recommendations on product design in terms of circularity performance (Shevchenko and Cluzel 2023). **Design quality improvement:** In connection to the Quality Management discipline, DPP allows designers to use the data originated from all the phases of the product life cycle to optimise not only the product design but also its production, use, and disposal (Adisorn *et al.* 2021).

3. PROPOSED FRAMEWORK FOR CIRCULAR DESIGN

To implement true and more efficient circularity there is the need for a circular product design methods that will allow the efficient remanufacturing of the products in their next life cycle. In the current section a proposed framework will be developed and presented to set the foundations and assist

during the product design phase for higher levels of circularity and by extent sustainability. The majority of the products are designed for one life cycle and then the next step is the recycling. This process creates a huge amount of waste because most of the products to be recycled have significant remaining useful life, and can still operate as products so when recycling them only the materials can be recovered and all the rest non-material such as time, manufacturing cost, human effort etc. Figure 1 represents this inefficient process, which is the driver of the proposed framework. The scope of the circular product design the non-materialized wastes will be minimized or ideally turned to zero. Therefore, companies need to design their products with a strategy and protocols for product re-manufacturing, upgrades/extensions, and potential re-use by other customers or re-purposing into other contexts. This strategy is vital for extending the usage and lifecycle(s) of the product, and if the product is not delivering the wanted value then re-purposing the product or re-using working components are options to consider. Down-cycling of the product is a further option in contexts with lower requirements.

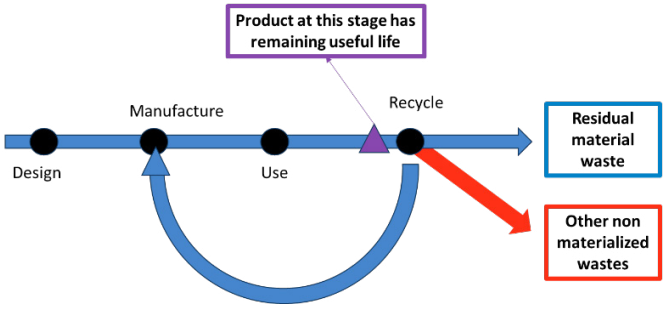


Figure 1. Current state of circular approaches

Circular product design is a more complex procedure than regular product design. In circular design there are many additional aspects that need to be taken into account when designing. Currently, in the product design there are only the actual product design and the manufacturing plan for this product. In order to apply more circular and sustainable approaches there is the need for additional aspect such as maintainability/extendibility/upgradability, disassembly plan, re-using, re-purposing, re-manufacturing and recycling plans during the product design process to be able to adjust the product design to fit as good as possible to all those aspects. Figure 2 illustrate the seven aspects of integrated circular product design identified by the authors. Integrated design is used as is an all-encompassing, holistic approach to design that combines specialisations that are typically thought of as distinct disciplines. It makes an effort to account for every variable and factor required for a decision-making process (Tamura *et al.* 2023). With blue are the two aspects that currently are most commonly addressed and with green are the proposed aspects that are affecting the circularity of the product.

3.1 Traditional product design steps

As mentioned above the traditional product design steps concerns the actual product design, which includes the fulfillment of the functional requirements of the product. Additionally, once there are some candidate product designs that are fulfilling the product functional requirements then the manufacturing plan of each design is produced in order to select the design with the most efficient manufacturing plan. This means that besides the product functional requirements there are the manufacturing requirements. In the era of circularity those requirements are significantly more, and the product design procedure is becoming more complex.

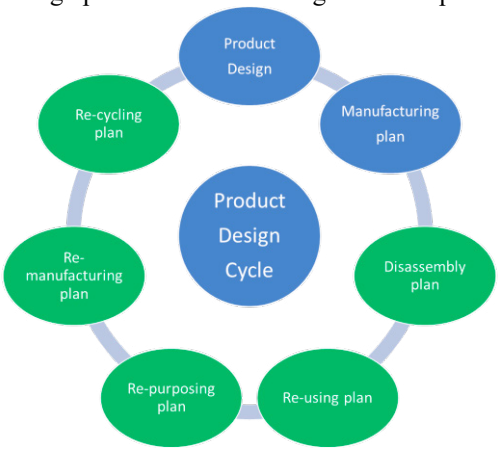


Figure 2. Aspects of circular product design

3.2 Circular product design aspects

In addition to the traditional product design aspects to implement circular approaches there is the need to consider circular requirements to the product design. The identified additional steps are collected and integrated to the proposed framework, from different research works on the remanufacturing domain. The first step after the manufacturing requirements are the reverse manufacturing requirements, which includes the disassembly plan of the product in case it is and assembly and not a single component (Sierra-Fontalvo *et al.* 2024). The disassembly plan is vital to be easy to perform because based on this the success of the circular approaches (product re-use, re-purpose, re-manufacturing and re-cycling) depends on it. Once the product ends the first or any of its potential following lifecycles, then its quality is evaluated and based on this quality inspection the right plan is selected, but regardless the selection the product needs to be disassembled at least to some extent to repair, remanufacture or upgrade (Geist and Balle 2024). Disassembly has to do also with the manufacturing and assembly characteristics, therefore in CE there is the need not only to plan for manufacturing and assembly, but also for efficient disassembly (Zhou and Xiao 2024).

Product re-using is the process where the product is being re-used as the same product with only some repairs or upgrades. In that way the remaining useful life of the entire product can be exploited near to 100% (Psarommatis and May 2022). In order to perform the necessary repairs and upgrades in an efficient and sustainable manner the product must be able to be easily diagnosed and disassembled. Therefore, the re-using plan should be include the most common actions for the re-use of a product. It is imperative that during the product design all the necessary protocols and procedures for the re-use are defined and the potential users of the product are educated toward this direction. During the product re-use there is a

possibility that the quality of the product will downgrade, therefore, with in the re-use plan there is the need to define the market where the re-use product will be available in order when the time comes the manufacturer to be ready to implement product re-use efficiently.

Product re-purpose is the process where the product is not fulfilling its purpose and cannot be re-use therefore, it is completely disassembled and the operational components can be re-used for the same functionality originally used or for any other functionality (Psarommatis and May 2022). The re-purposing process is more complicated than the re-use as there are many more processes to be done and applications that the components can be re-used. The re-purposing plan should include not only the procedures for accomplishing the re-purposing, but also the different potential markets that the re-purposed components with be used. At the same time the design of the product should consider the use of re-purposed or re-used components from the available in the market. It is imperative for the success of the circular approaches that not only consider how to re-use re-purpose of products but also where those products or components can be used. To this extent there is the need of collaborative product design between different industries in order to make sure that the designed product can fulfill both the first life cycle requirements but also the X life cycle requirements.

For the recycling the same process described before applies. More specifically, during the design of the product there is the need to define the recycling requirements for the specific product in order to use the proper materials and manufacturing methods to make recycling easier according to the recycling requirements of the corresponding time. As these changes may be costly and require new competences/knowledge, all manufacturing companies might not survive the transition whereas others may find new opportunities together with fast moving entrants to markets. Thus, some jobs will likely be lost whereas new created.

3.3 Legal aspect of circularity

Further, as more and more products have considerable software constituents and are connected to networks and Internet, there are existing and emerging legal requirements to be able to maintain, update and upgrade the cybersecurity features i.e., software, communication protocols, platforms, hardware as well as necessary organizational structures) throughout the whole lifecycle (c.f. EU's GDPR, Cyber Resilience Act (EU Cyber Resilience Act | Shaping Europe's digital future 2024), RED (Radio Equipment Directive (RED) - European Commission 2024), Directive on the Resilience of Critical Entities (Critical infrastructure resilience - European Commission 2024), or NIS2 (NIS Directive 2024), etc.). The legal requirements differ around the globe, if personal data is transmitted, and which contexts the product are to be used within. There is a large risk that the increasingly sophisticated number of cyberattacks will make unsecure products unsellable and unusable if cybersecurity is not catered for during the whole lifecycle as well as the concept of Secure Development Lifecycle (Good Practices for Security of IoT - Secure Software Development Lifecycle 2024) (SDLC) is used during the design. Further, health and safety requirements must also be considered during the design (Regulation 2023/1230/EU - machinery | Safety and health at work EU-OSHA 2024). An expected consequence of increased cybersecurity in software is that connected products will become more costly to procure as well as to maintain and support, which the buyers/customers will have to pay for.

3.4. Novelty of proposed framework for product design cycle

The framework couples traditional, circular, and legal aspects of product design into a lean and easy to understand cycle (see Figure 2) for product designers. The unusually fast addition of recent and expected legal and regulatory compliance requirements, will cast a large burden upon product designers who must work together with compliance teams. These together need to consider this as part of the product design and, in particular, the wanted longer usage/operations phase including maintenance of products' life cycles. Else, this will cost a lot to fix later or might even make poorly designed products unsellable or unmanageable later in the life cycle.

3.5. Examples

Plastic bags when grocery shopping example. Plastic bags, which commonly are bought and used in grocery stores to transport home the groceries, can be of low quality and barely hold together until the home is reached or be of high quality enabling to be re-used a number of times. The common end-of-life for a plastic bag is either that it is disposed of in nature or more responsibly used a container for plastic packages and together recycled at plastic recycling stations or being burnt up as a container for common household trash. However, a better option to "simple" plastic bags is to use durable multi-time use grocery shopping bags made out of plastic (or organic fibers) that also are re-cyclable. In this example, design-for-durability and re-cyclability combined with knowledge transfer to users and product information provided can be key together with the product design (and the plastic material choice), and plans for re-use/re-purpose/re-cycling (see Figure 2).

Battery packs of different equipment. Batteries are a product that is well established in our modern lives, from mobile phones, cars, home appliances, hand tools etc. Additionally, such product is considered one of the products with the highest environmental footprint (Gutsch and Leker 2024). The literature currently states the difficulty of recycling batteries and a lot of effort is done for the remanufacturing of batteries, but the biggest difficulty is due to the design of the batteries. Usually, batteries can be found in the form of a battery packs, which means that the batteries are assembled with other components. When those battery packs comes to the end of first life might be for many reasons, either the casing is damaged or the batteries are not performing well, which means that when recycled components with remaining useful life are recycled as well. Due to the way the battery packs are designed make very difficult and inefficient their remanufacturing. In such case the proposed circular design could significantly improve the environmental footprint of battery packs

4. DISCUSSION

To decrease the ecologic impact and footprint of products, is to extend the use-phase. However, this requires a design with

long-term circular planning, adaptability and change management. A number of eco-innovations and eco-efficiency practices such as eco-design and circular design. While important depending on the context, legal and regulatory requirements, as well as company accountability, such practices are insufficient in delivering a holistic change.

During recent years, in particular within the EU, the legal and regulatory requirements pertaining to product and services provided, has accelerated and now or soon need to be part of product design. Further, as these legal and regulatory requirements slowly get fairly harmonized across the globe, product design need to address future adaptability and changeability of products.

Depending on targeted use context for products, health and safety are further important design issues to not get an unplanned and abrupt stop of the lifecycle. Unfortunately, cybersecurity requirements for the intended use context also need to be managed as part of the design and for instance EU has existing and emerging legislation and regulations to tackle the increased number of cyberattacks costing society a lot of money as well as unavailability of critical services. Thus, for instance connected products need to be able to be updated throughout the lifecycle, which will require that the provider develops and operate new processes, structures, and also educate/train their personnel on how to develop, support, and operate cybersecure (connected) products and/or services.

If the product is designed, developed, and supported, etc., by a collaborative value-chain, it must be clear which part/actor of the value-chain who has the legal product responsibility as well as other necessary responsibilities and how the value is captured by the different actors in the value-chain. All involved actors need to have a win situation including the user/customer, or else the product lifecycle will likely end.

A key technology for the successful implementation of circularity if the DPP. All the plans and design requirements, protocols and procedures need to be documented to the product's DPP in order to be used in the future from the different stakeholders in the value chain.

5. CONCLUSIONS

The paper contributes to literature by proposing a lean and easy to understand framework for achieving sustainable manufacturing of products through coupling traditional, circular and legal aspect as part of the product design. Product design used to include only the actual product design and the manufacturing plan. To achieve circularity there is a need to add more aspects to the product design that are coming from the concept of circular economy. Therefore, we have defined additional design steps that are adding more aspects. Those steps include the incorporation of the product re-use, repurpose, re-manufacture and re-cycling. To achieve advanced circularity, needed is to design the product based on all those aspects. However, this will make the design process further complex, but it will significantly assist to increase the sustainability and mitigate the increased product needs from the market. Further, the paper contributes to practice by indicating that there will be change, and to be successful, the change must be embraced and prepared for. The manufacturing companies who will not embrace will get passed by those able to change or new fast entrants to markets.

In addition, the managerial contribution is that the paper highlights the need for a transition where new supporting structures, processes and competences/knowledge is needed. Thus, investments, changes and potential acquiring of necessary competences/knowledge must be catered for.

The proposed approach can serve as a viable method for integrated circular product design, in the current approach focused on the methodological aspect but the legal aspect and intellectual property aspects were not considered as they would make the approach overcomplicated at this stage. Future research should focus on the incorporation of those aspects to the proposed framework.

ACKNOWLEDGEMENTS

The presented work was partially supported by the projects CIRCULess and PLOOTO, EU H2020 projects under grant agreements No 101138330 and 101092008.

REFERENCES

Acerbi, F., Sassanelli, C., and Taisch, M., 2022. A conceptual data model promoting data-driven circular manufacturing. *Operations Management Research*, 15 (3), 838–857.

Adisorn, T., Tholen, L., and Götz, T., 2021. Towards a Digital Product Passport Fit for Contributing to a Circular Economy. *Energies*, 14 (8), 2289.

Bocken, N.M.P., Short, S.W., Rana, P., and Evans, S., 2014. A literature and practice review to develop sustainable business model archetypes. *Journal of Cleaner Production*, 65, 42–56.

Cong, L., Zhao, F., and Sutherland, J.W., 2019. A Design Method to Improve End-of-Use Product Value Recovery for Circular Economy. *Journal of Mechanical Design*, 141 (044502).

Critical infrastructure resilience - European Commission [online], 2024. Available from: https://home-affairs.ec.europa.eu/policies/internal-security/counter-terrorism-and-radicalisation/protection/critical-infrastructure-resilience_en [Accessed 15 Jan 2024].

EU Cyber Resilience Act | Shaping Europe's digital future [online], 2024. Available from: https://digital-strategy.ec.europa.eu/en/policies/cyber-resilience-act [Accessed 15 Jan 2024].

Geist, H. and Balle, F., 2024. Remanufactured products, components, and their materials: A circularity engineering focused empirical status quo analysis. *Sustainable Production and Consumption*, 45, 525–537.

Good Practices for Security of IoT - Secure Software Development Lifecycle [online], 2024. *ENISA*. Available from:

https://www.enisa.europa.eu/publications/good-practices-for-security-of-iot-1 [Accessed 15 Jan 2024].

Guo, X., Zhao, W., Hu, H., Li, L., Liu, Y., Wang, J., and Zhang, K., 2021. A smart knowledge deployment method for the conceptual design of low-carbon products. *Journal of Cleaner Production*, 321, 128994.

Gutsch, M. and Leker, J., 2024. Costs, carbon footprint, and environmental impacts of lithium-ion batteries – From cathode active material synthesis to cell manufacturing and recycling. *Applied Energy*, 353, 122132.

Industry 5.0 | European Commission [online], 2021. Available from: https://ec.europa.eu/info/research-and-innovation/research-area/industrial-research-and-innovation/industry-50_en [Accessed 16 Nov 2021].

Jabbour, C.J.C., Colasante, A., D'Adamo, I., Rosa, P., and Sassanelli, C., 2023. Customer Attitudes Toward Circular Economy in the E-Waste Context: A Survey Assessing Sustainable Consumption Dynamics. *IEEE Engineering Management Review*, 51 (4), 28–45.

Jansen, M., Meisen, T., Plociennik, C., Berg, H., Pomp, A., and Windholz, W., 2023. Stop Guessing in the Dark: Identified Requirements for Digital Product Passport Systems. *Systems*, 11 (3), 123.

K.e.k, V., Kandasamy, J., and Gite, V., 2021. A framework to assess circularity across product-life cycle stages – A case study. *Procedia CIRP*, 98, 442–447.

King, M.R.N., Timms, P.D., and Mountney, S., 2023. A proposed universal definition of a Digital Product Passport Ecosystem (DPPE): Worldviews, discrete capabilities, stakeholder requirements and concerns. *Journal of Cleaner Production*, 384, 135538.

McKendry, D.A., Whitfield, R.I., and Duffy, A.H.B., 2022. Product Lifecycle Management implementation for high value Engineering to Order programmes: An informational perspective. *Journal of Industrial Information Integration*, 26, 100264.

NIS Directive [online], 2024. *ENISA*. Available from: https://www.enisa.europa.eu/topics/cybersecurity-policy/nis-directive-new [Accessed 15 Jan 2024].

Psarommatis, F. and May, G., 2022. Achieving Global Sustainability Through Sustainable Product Life Cycle. *IFIP Advances in Information and Communication Technology*, 663 IFIP, 391–398.

Psarommatis, F. and May, G., 2024. Digital Product Passport: A Pathway to Circularity and Sustainability in Modern Manufacturing. *Sustainability*, 16 (1), 396.

Radio Equipment Directive (RED) - European Commission [online], 2024. Available from: https://single-market-economy.ec.europa.eu/sectors/electrical-and-electronic-engineering-industries-eei/radio-equipment-directive-red_en [Accessed 15 Jan 2024].

Regulation 2023/1230/EU - machinery | Safety and health at work EU-OSHA [online], 2024. Available from: https://osha.europa.eu/en/legislation/directive/regulation-20231230eu-machinery [Accessed 15 Jan 2024].

Rentizelas, A. and Trivyza, N.L., 2022. Enhancing circularity in the car sharing industry: Reverse supply chain network design optimisation for reusable car frames. *Sustainable Production and Consumption*, 32, 863–879.

Romani, A., Rognoli, V., and Levi, M., 2021. Design, Materials, and Extrusion-Based Additive Manufacturing in Circular Economy Contexts: From Waste to New Products. *Sustainability*, 13 (13), 7269.

Sassanelli, C., Urbinati, A., Rosa, P., Chiaroni, D., and Terzi, S., 2020. Addressing circular economy through design for X approaches: A systematic literature review. *Computers in Industry*, 120, 103245.

Shevchenko, T. and Cluzel, F., 2023. A CIRCULAR PRODUCT DESIGN TOOLKIT: METHODOLOGICAL BACKGROUND, BASIC PRINCIPLES AND BUILDING BLOCKS. *Proceedings of the Design Society*, 3, 59–68.

Sierra-Fontalvo, L., Polo-Cardozo, J., Maury-Ramírez, H., and Mesa, J.A., 2024. Diagnosing remanufacture potential at product-component level: A disassemblability and integrity approach. *Resources, Conservation and Recycling*, 205, 107529.

Tamura, T., Odagaki, R., Umeda, Y., Miyake, G., Matsuda, G., Tajima, A., and Kishita, Y., 2023. Proposal of an integrated design framework for ICT-supported Circular Economy business. *Procedia CIRP*, 116, 564–569.

Terzi, S., Bouras, A., Dutta, D., Garetti, M., and Dimitris Kiritsis, 2010. Product lifecycle management – from its history to its new role. *International Journal of Product Lifecycle Management*, 4 (4), 360–389.

Tolio, T., Bernard, A., Colledani, M., Kara, S., Seliger, G., Duflou, J., Battaia, O., and Takata, S., 2017. Design, management and control of demanufacturing and remanufacturing systems. *CIRP Annals*, 66 (2), 585–609.

Ulrich, K.T. and Eppinger, S.D., 2016. *Product design and development*. McGraw-Hill.

Urbinati, A., Franzò, S., Sassanelli, C., Rosa, P., Chiaroni, D., and Terzi, S., 2023. Exploiting 3-D Printing for Designing Circular Business Models: A Novel Framework. *IEEE Transactions on Engineering Management*, 1–15.

Zhou, X. and Xiao, C., 2024. Maintenance modeling for multi-station remanufacturing systems with interaction between production schedule and station reliability. *CIRP Journal of Manufacturing Science and Technology*, 50, 113–126.